

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Assignment 01



Submitted to:

Sir Hassan Shah

Submitted by:

Zulqarnain Hassan

Roll No: SP21379

Subject: information security

Department: BSSE-6-Afternoon

Date: 24-3-2025

Day: _____

Date: _____

1. Substitution:

Plain text = ZULQARNAIN

cipher = 2621 12 17 01 17 14 01 09 14

Rot 13

Plain text = ZULQARNAIN

A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Cipher text = M H Y D N E A N V A

xOR

Plai text = ~~ZULQ~~ ZULQARNAIN
Combinator AMBER JACKS

$Z \oplus A$

$$\begin{array}{r} 11010 \\ 00001 \\ \hline 11011 = 27 \end{array}$$

$U \oplus M$

$$\begin{array}{r} 10101 \\ 01101 \\ \hline 11000 = 24 \end{array}$$

Day: _____

Date: _____

$L \oplus B$

$$\begin{array}{r} 1100 \\ 0010 \\ \hline 1110 = 14 \end{array}$$

$Q \oplus F$

$$\begin{array}{r} 10001 \\ 00011 \\ \hline 10010 = 18 \end{array}$$

$A \oplus R$

$$\begin{array}{r} 00001 \\ 10010 \\ \hline 10011 = 19 \end{array}$$

$R \oplus J$

$$\begin{array}{r} 10010 \\ 01010 \\ \hline 11000 = 24 \end{array}$$

$N \oplus A$

$$\begin{array}{r} 1110 \\ 0001 \\ \hline 1101 = 13 \end{array}$$

$A \oplus C$

$$\begin{array}{r} 11 \\ 01 \\ \hline 10 = 2 \end{array}$$

$I \oplus K$

$$\begin{array}{r} 1001 \\ 1011 \\ \hline 0010 = 2 \end{array}$$

Day: _____

Date: _____ 3

NBS

synopsis

1110

10011

11101 = 29

Cipher = 27241418192415020229

Ceaser:

Plain text = ZULQARNAIN

cipher text = CXOTDUQDMQ

Affine

$a = 5, b = 9 \quad (ax+b) \% 26$

Plain text	Z	U	L	Q	A	R	N	A	I	N
x	25	20	11	16	0	17	13	0	8	13
$(ax+b) \% 26$	4	5	12	11	9	16	12	9	23	12
Cipher text	E	F	M	L	J	Q	M	J	X	M

Cipher text = FUMLJQM JXM

Day: _____

Date: _____

4

Vigenere

Key = SECURE

TEXT	Z	U	L	Q	A	R	N	A	I	N
P	25	20	11	16	0	17	13	0	8	13
Key										
Key	S	E	C	U	R	E	S	E	C	U
K	18	4	2	20	17	4	18	4	2	20
$(P+K) \% 26$	6	23	12	9	16	20	4	3	9	6
	G	X	M	J	Q	U	E	D	J	G

cipher text = GXMJQUEDJG

Transpositional

Plaintext = ZULQARNAIN

Z				A				I	
	U		Q		R		A		N
		L				N			

cipher text = ZAIUQARANIN

Day: _____

Date: _____

One Time Pad

Plaintext = ~~ZULBARNAIN~~ ZULBARNAIN

Pad = NUML

Text	Pad	Text+Pad-1	Cipher
Z 26	N 14	14	N
U 21	U 21	16	O
L 12	M 13	24	X
B 17	L 12	2	B
A 1	N 14	14	N
R 18	U 21	12	L
N 14	M 13	26	Z
A 1	L 12	12	L
I 9	N 14	22	V
N 14	U 21	8	H

Cipher text = NOXBNLZLVH